

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1254

Vapor 138 kV
May 16, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1254. The Interconnection Customer proposes a 118.9 MW solar photovoltaic facility interconnecting at the Vapor 138 kV substation in the Pennsylvania area. The request was received on July 20, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 346 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,719,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Cadillac Renewable Energy / Grs Arbor Hills 138kV	Simpson Stuart Tap 345-138 kV Transformer Bank	113.6	209	15.2
Lewisville / Williams 138kV L	Us Gypsum / Forbes 138kV	112.0	253	5.3
Grs Arbor Hills / Firstenergy Mitchell Power Station 138kV	Simpso Stuart Tap 345/138 kV Transformer Bank	102.3	222	15.2
Matanzas / John Day 138kV	Kempster / Simpson Stuart Tap 138kV	102.1	337	5.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Grs Arbor Hills / Cadillac Renewable Energy 138kV	Rebuild line with higher-capacity conductor	\$4,893,000	44% (\$2,143,000)
Lewisville / Williams 138kV L	Rebuild line with higher-capacity conductor	\$1,540,000	3% (\$53,000)
Grs Arbor Hills / Firstenergy Mitchell Power Sta 138kV line	Reconductor line	\$949,000	44% (\$416,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Matanzas / John Day 138kV	Reconductor line	\$637,000	17% (\$107,000)
	Total allocated to project		\$2,719,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.